THIS IS AN OPEN BOOK EXAMINATION
(INTERNET SEARCHING IS NOT ALLOWED)
YOU MUST HAND-WRITE YOUR ANSWER

THE UNIVERSITY OF HONG KONG
DEPARTMENT OF CHEMISTRY

CHEM1043 General Chemistry II

May 14, 2022 (Saturday) **9:30 am – 11:30 am**

(Cutoff time for uploading your answer script to OLEX: **11:45 am**)

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Candidates are permitted to refer to the following electronic/printed/handwritten materials in the examination: textbook (or reference book), lecture slides, assignment handout, and self-made notes. Internet searching and crowdsourcing from group messages, online forums or social media, etc. are strictly forbidden.

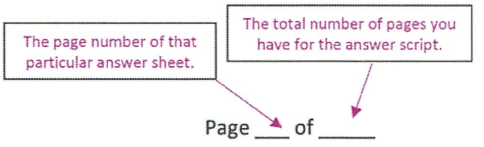
This paper consists of **six** questions (100 marks). Answer **ALL** questions.
(Q1: 12 marks, Q2: 18 marks, Q3: 17 marks, Q4: 17 marks, Q5: 17 marks, Q6: 19 marks)

Hand-write your answers on A4 size white paper using black pen or blue pen.
Write on each answer sheet your student number and the page number.

A list of physical constants and a periodic table are provided on the last pages of this question paper.

At the end of the examination, please do the following:

- ☐ Check that you have written your **student number** and **page number** on each page of your answer script.
- ☐ Scan or take photos of your answer script and combine the pages into one single file.
- ☐ Check your file to make sure all pages are included and the images are clear.
- ☐ Submit your file through OLEX before the cutoff time. And **make sure you have submitted the correct and complete file**.



There are **six questions** in this paper. Answer **ALL** questions. **Hand-write** your answers on A4 paper.

Question 1 (12 marks)

(a) A 2.650 g sample of a gaseous compound occupies 428 mL at 24.3 °C and 742 mmHg. The compound consists of 15.5% C, 23.0% Cl, and 61.5% F, by mass. What is its molecular formula? Show your calculations.

(5 marks)

(b) A 3.57 g sample of a KCl–$KClO_3$ mixture is decomposed by heating and produces 119 mL O_2 (g) measured at 22.4 °C and 98.3 kPa. What is the mass percent of $KClO_3$ in the mixture? Show your calculations.

$$2\,KClO_3(s) \rightarrow 2\,KCl(s) + 3\,O_2(g)$$

(4 marks)

(c) If 1×10^{22} N_2 molecules effuse through a pinhole in 60 s, what amount of H_2 gas (in mol) would effuse through the same pinhole in 60 s? Show your calculations.

(3 marks)

Question 2 (18 marks)

(a) Explain the essential difference in how the valence bond method and molecular orbital theory describe a covalent bond.

(6 marks)

(b) Consider the ions NO^+ and N_2^+ and use molecular orbital theory to answer the following:

(i) Write the molecular orbital diagram and also the configuration of each ion (ignore the 1s electrons). (6 marks)

(ii) Predict the bond order of each ion. (2 marks)

(iii) Which of these ions is paramagnetic? Which is diamagnetic? Explain your answer. (2 marks)

(iv) Which of these ions do you think has the greater bond length? Explain. (2 marks)

Question 3 (17 marks)

(a) At 25 °C and 0% salinity the amount of oxygen in the ocean is 5.77 mL/L. What is the molarity of oxygen in ocean water? (2 marks)

(b) The aqueous solubility of $CO_2(g)$ at 20 °C and 1.00 atm is equivalent to 87.8 mL $CO_2(g)$ measured at 0 °C and 1.00 atm, per 100 mL of water. What is the molarity of $CO_2(g)$ in water that is at 20 °C and saturated with air at 1.00 atm? The volume percent of $CO_2(g)$ in air is 0.0360%. Show your calculations.
[Assume that the volume of the water does not change when it becomes saturated with air.]
(4 marks)

(c) Assume complete dissociation for the following aqueous solutions. Rank them from the lowest to the highest freezing point. Explain your answer.

$0.10\ mol\ kg^{-1}\ FeCl_3$, $0.30\ mol\ kg^{-1}$ urea, $0.15\ mol\ kg^{-1}\ MgCl_2$

(6 marks)

(d) What is the vapour pressure in mmHg of a solution that is prepared by dissolving 7.50 g of benzoic acid (C_6H_5COOH) in 150.00 g of ethanol (CH_3CH_2OH) at 35 °C? The vapour pressure of pure ethanol at 35 °C is 100.5 mmHg. Show your calculations.
(5 marks)

Question 4 (17 marks)

(a) A solution contains 0.020 M Cl^- ions and 0.020 M Br^- ions. To separate the Cl^- ions from the Br^- ions, solid $AgNO_3$ is slowly added to the solution. What concentration of Ag^+ ions is needed to precipitate as much AgBr as possible without precipitating AgCl? Show your calculations and reasoning.
[Assume volume remains unchanged upon the addition of solid $AgNO_3$ to the solution and take K_{sp} of AgCl = 1.8×10^{-10} and K_{sp} of AgBr = 5.0×10^{-13}.]
(7 marks)

(b) Which compound, CuS or AgCl, will be more soluble in acidic solution than in water? Explain your answer.
(5 marks)

(c) 0.20 mol of $CuSO_4$ is added to a litre of 1.20 M NH_3 solution. What is the concentration of Cu^{2+} ions at equilibrium? Given K_f of $[Cu(NH_3)_4]^{2+} = 1.1 \times 10^{13}$. Show the steps by which you arrive at the answer.
(5 marks)

Question 5 (17 marks)

(a) 2.00 mol of an ideal gas at 25 °C expands isothermally and reversibly from 9.00 L to 36.00 L. Calculate q, w, ΔU and ΔH for the process. Show your calculations.

(6 marks)

(b) Calculate the entropy change, ΔS, for each of the following processes. Show your calculations.

(i) A sample of two moles of Ne (g) undergoes an expansion from 25 L to 50 L at 25 °C.

(ii) The temperature of a sample of three moles of CH_4 (g) is increased from 25 °C to 50 °C at a constant pressure of 1 bar.

[The molar constant pressure heat capacities of Ne (g) and CH_4 (g) are 20.8 $J\,K^{-1}\,mol^{-1}$ and 35.7 $J\,K^{-1}\,mol^{-1}$, respectively.]

(5 marks)

(c) The equilibrium constant (K_p) for the reaction below is 0.113 at 298 K.

$$N_2O_4\,(g) \rightleftharpoons 2NO_2\,(g)$$

In an experiment, the initial pressures are $P_{NO_2} = 0.122$ bar and $P_{N_2O_4} = 0.453$ bar. Using the given data, calculate the standard free energy of reaction ($\Delta_r G°$) and the free energy of reaction ($\Delta_r G$) for this decomposition reaction of N_2O_4 to NO_2 at these pressures. Hence, predict the direction of the reaction under the given initial conditions toward equilibrium. Show your calculations and reasoning.

(6 marks)

Question 6 (19 marks)

(a) (i) Sketch a diagram to show a voltaic cell consisting of one half-cell with a Cr electrode in a $Cr(NO_3)_3$ solution, another half-cell with an Ag electrode in an $AgNO_3$ solution, and any other features necessary to complete the cell.

(3 marks)

(ii) For the voltaic cell described in (i), give the half-cell equations and the overall equation.

(2 marks)

(iii) Use data from **Table 1** (on P. 6) to determine $E°_{cell}$ for this voltaic cell. Show your calculations. Hence, identify the anode and cathode, and give the direction of electron flow in this voltaic cell.

(4 marks)

(b) Use data in **Table 1** (on P. 6) to calculate $\Delta_r G°$ for the following reaction. Hence, comment on the spontaneity of the reaction. Show your calculations and reasoning.

$$Cu^{2+}(aq) + Fe(s) \rightarrow Cu(s) + Fe^{2+}(aq)$$

(5 marks)

(c) Consider the following cell.

$$Ag(s) \,|\, Ag^{+}(0.090\ M) \,||\, Hg^{2+}(0.75\ M) \,|\, Hg(l)$$

Will the cell reaction proceed spontaneously? Show your calculations and reasoning.

(5 marks)

– END OF PAPER –

(Table 1) Standard Reduction Potentials at 298.15 K and 1 atm

Reduction Half-Cell Reaction	Standard Reduction Potential, V
Acidic solution	
$F_2(g) + 2\,e^- \rightleftharpoons 2\,F^-(aq)$	+2.866
$O_3(g) + 2\,H^+(aq) + 2\,e^- \rightleftharpoons O_2(g) + H_2O(l)$	+2.075
$S_2O_8^{2-}(aq) + 2\,e^- \rightleftharpoons 2\,SO_4^{2-}(aq)$	+2.01
$H_2O_2(aq) + 2\,H^+(aq) + 2\,e^- \rightleftharpoons 2\,H_2O(l)$	+1.763
$Ce^{4+}(aq) + e^- \rightleftharpoons Ce^{3+}(aq)$	+1.72
$MnO_4^-(aq) + 4\,H^+(aq) + 3\,e^- \rightleftharpoons MnO_2(s) + 2\,H_2O(l)$	+1.70
$PbO_2(s) + SO_4^{2-}(aq) + 4H^+(aq) + 2e^- \rightleftharpoons PbSO_4(s) + 2\,H_2O(l)$	+1.69
$Cl_2(g) + 2\,e^- \rightleftharpoons 2\,Cl^-(aq)$	+1.36
$Cr_2O_7^{2-}(aq) + 14\,H^+(aq) + 6\,e^- \rightleftharpoons 2\,Cr^{3+}(aq) + 7\,H_2O(l)$	+1.32
$MnO_2(s) + 4\,H^+(aq) + 2\,e^- \rightleftharpoons Mn^{2+}(aq) + 2\,H_2O(l)$	+1.23
$MnO_4^-(aq) + 8\,H^+(aq) + 5\,e^- \rightleftharpoons Mn^{2+}(aq) + 4\,H_2O(l)$	+1.51
$Cu^{2+}(aq) + 2\,e^- \rightleftharpoons Cu(s)$	+0.340
$O_2(g) + 4\,H^+(aq) + 4\,e^- \rightleftharpoons 2\,H_2O(l)$	+1.229
$IO_3^-(aq) + 12\,H^+(aq) + 10\,e^- \rightleftharpoons I_2(s) + 6\,H_2O(l)$	+1.20
$Br_2(l) + 2\,e^- \rightleftharpoons 2\,Br^-$	+1.065
$NO_3^-(aq) + 4\,H^+(aq) + 3\,e^- \rightleftharpoons NO(g) + 2\,H_2O(l)$	+0.956
$2\,Hg^{2+}(aq) + 2\,e^- \rightleftharpoons Hg(l)$	+0.854
$Ag^+(aq) + e^- \rightleftharpoons Ag(s)$	+0.800
$Fe^{3+}(aq) + e^- \rightleftharpoons Fe^{2+}(aq)$	+0.771
$O_2(g) + 2\,H^+(aq) + 2\,e^- \rightleftharpoons H_2O_2(aq)$	+0.695
$I_2(s) + 2\,e^- \rightleftharpoons 2\,I^-(aq)$	+0.535
$Cu^+(aq) + e^- \rightleftharpoons Cu(s)$	+0.520
$SO_4^{2-}(aq) + 4\,H^+(aq) + 2\,e^- \rightleftharpoons 2\,H_2O(l) + SO_2(g)$	+0.17
$Sn^{4+}(aq) + 2\,e^- \rightleftharpoons Sn^{2+}(aq)$	+0.154
$S(s) + 2\,H^+(aq) + 2\,e^- \rightleftharpoons H_2S(g)$	+0.144
$2\,H^+(aq) + 2\,e^- \rightleftharpoons H_2(g)$	0.0
$Pb^{2+}(aq) + 2\,e^- \rightleftharpoons Pb(s)$	−0.125
$Sn^{2+}(aq) + 2\,e^- \rightleftharpoons Sn(s)$	−0.137
$Fe^{2+}(aq) + 2\,e^- \rightleftharpoons Fe(s)$	−0.440
$Zn^{2+}(aq) + 2\,e^- \rightleftharpoons Zn(s)$	−0.763
$Cr^{3+}(aq) + 3\,e^- \rightleftharpoons Cr(s)$	−0.740
$Al^{3+}(aq) + 3\,e^- \rightleftharpoons Al(s)$	−1.662
$Mg^{2+}(aq) + 2\,e^- \rightleftharpoons Mg(s)$	−2.372
$Na^+(aq) + e^- \rightleftharpoons Na(s)$	−2.714
$Ca^{2+}(aq) + 2\,e^- \rightleftharpoons Ca(s)$	−2.868
$K^+(aq) + e^- \rightleftharpoons K(s)$	−2.931
$Li^+(aq) + e^- \rightleftharpoons Li(s)$	−3.05
Basic solution	
$O_3(g) + H_2O(l) + 2\,e^- \rightleftharpoons O_2(g) + 2\,OH^-(aq)$	+1.246
$OCl^-(aq) + H_2O(l) + 2\,e^- \rightleftharpoons Cl^-(aq) + 2\,OH^-(aq)$	+0.890
$O_2(g) + 2\,H_2O(l) + 4\,e^- \rightleftharpoons 4\,OH^-(aq)$	+0.401
$2\,H_2O(l) + 2\,e^- \rightleftharpoons H_2(g) + 2\,OH^-(aq)$	−0.828

Physical Constants

Planck's constant	$h = 6.626 \times 10^{-34}$ J s molecule^{-1}
Velocity of light in vacuum	$c = 2.998 \times 10^{8}$ m s^{-1}
Avogadro's number	$N = 6.022 \times 10^{23}$ molecules mol^{-1}
Boltzmann constant	$k = 1.381 \times 10^{-23}$ J K^{-1} molecule^{-1}
Gas constant	$R = 8.314$ J K^{-1} mol^{-1} $= 0.08206$ dm^3 atm K^{-1} mol^{-1} $= 0.08314$ dm^3 bar K^{-1} mol^{-1} $= 8.314$ m^3 Pa K^{-1} mol^{-1}
Faraday constant	$F = 96487$ C mol^{-1} (of electrons transferred) $= 23061$ cal volt^{-1} mol^{-1} (of electrons transferred)
Elementary charge	$e = 1.602 \times 10^{-19}$ C
Mass of electron	$m_e = 9.109 \times 10^{-31}$ kg
Mass of proton	$m_p = 1.673 \times 10^{-27}$ kg
Atomic mass constant	$m_u = 1.661 \times 10^{-27}$ kg
Standard gravitational acceleration	$g = 9.807$ m s^{-2}

Conversion Tables:	Numerical Constants:
1 cal = 4.184 J (exactly) 1 erg = 10^{-7} J 1 G = 10^{-4} T (G = gauss, T = tesla) 1 atm = 101325 Pa = 760 mmHg (Torr) 1 bar = 10^5 Pa 0 °C = 273.15 K	π = 3.141593 e = 2.718282 ln x = 2.303 log x

Energy Conversion Factors:				
	cal mol^{-1}	J molecule^{-1}	eV molecule^{-1}	cm^{-1}
1 cal mol^{-1} =	1	6.947×10^{-24}	4.336×10^{-5}	0.3500
1 J molecule^{-1} =	1.439×10^{23}	1	6.242×10^{18}	5.036×10^{22}
1 eV molecule^{-1} =	2.306×10^{4}	1.602×10^{-19}	1	8.068×10^{3}
1 cm^{-1} =	2.859	1.986×10^{-23}	1.240×10^{-4}	1

Some Equations for Kinetic Theory of Gases and Electrochemistry:

$u_{rms} = \sqrt{\frac{3RT}{M}}$	$\overline{E_k} = \frac{3}{2}\left(\frac{RT}{N_A}\right)$	$P = \frac{1}{3}\frac{N}{V}\overline{mu^2}$
$\Delta_r G^\circ = -zFE^\circ_{cell}$	$E^\circ_{cell} = \frac{0.025693\ V}{z} \ln K$	$E_{cell} = E^\circ_{cell} - \frac{0.0257\ V}{z} \ln Q$

The periodic table

1	2		3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Hydrogen 1 **H** 1.008																		Helium 2 **He** 4.0026
Lithium 3 **Li** 6.94	Beryllium 4 **Be** 9.0122												Boron 5 **B** 10.81	Carbon 6 **C** 12.011	Nitrogen 7 **N** 14.007	Oxygen 8 **O** 15.999	Fluorine 9 **F** 18.998	Neon 10 **Ne** 20.180
Sodium 11 **Na** 22.990	Magnesium 12 **Mg** 24.305												Aluminium 13 **Al** 26.982	Silicon 14 **Si** 28.085	Phosphorus 15 **P** 30.974	Sulfur 16 **S** 32.06	Chlorine 17 **Cl** 35.45	Argon 18 **Ar** 39.948
Potassium 19 **K** 39.098	Calcium 20 **Ca** 40.078(4)		Scandium 21 **Sc** 44.956	Titanium 22 **Ti** 47.867	Vanadium 23 **V** 50.942	Chromium 24 **Cr** 51.996	Manganese 25 **Mn** 54.938	Iron 26 **Fe** 55.845(2)	Cobalt 27 **Co** 58.933	Nickel 28 **Ni** 58.693	Copper 29 **Cu** 63.546(3)	Zinc 30 **Zn** 65.38(2)	Gallium 31 **Ga** 69.723	Germanium 32 **Ge** 72.63	Arsenic 33 **As** 74.922	Selenium 34 **Se** 78.96(3)	Bromine 35 **Br** 79.904	Krypton 36 **Kr** 83.798(2)
Rubidium 37 **Rb** 85.468	Strontium 38 **Sr** 87.62		Yttrium 39 **Y** 88.906	Zirconium 40 **Zr** 91.224(2)	Niobium 41 **Nb** 92.906(2)	Molybdenum 42 **Mo** 95.96(2)	Technetium 43 **Tc** [97.91]	Ruthenium 44 **Ru** 101.07(2)	Rhodium 45 **Rh** 102.91	Palladium 46 **Pd** 106.42	Silver 47 **Ag** 107.87	Cadmium 48 **Cd** 112.41	Indium 49 **In** 114.82	Tin 50 **Sn** 118.71	Antimony 51 **Sb** 121.76	Tellurium 52 **Te** 127.60(3)	Iodine 53 **I** 126.90	Xenon 54 **Xe** 131.29
Caesium 55 **Cs** 132.91	Barium 56 **Ba** 137.33	57-70 *	Lutetium 71 **Lu** 174.97	Hafnium 72 **Hf** 178.49(2)	Tantalum 73 **Ta** 180.95	Tungsten 74 **W** 183.84	Rhenium 75 **Re** 186.21	Osmium 76 **Os** 190.23(2)	Iridium 77 **Ir** 192.22	Platinum 78 **Pt** 195.08	Gold 79 **Au** 196.97	Mercury 80 **Hg** 200.59	Thallium 81 **Tl** 204.38	Lead 82 **Pb** 207.2	Bismuth 83 **Bi** 208.98	Polonium 84 **Po** [208.98]	Astatine 85 **At** [209.99]	Radon 86 **Rn** [222.02]
Francium 87 **Fr** [223.02]	Radium 88 **Ra** [226.03]	89-102 **	Lawrencium 103 **Lr** [262.11]	Rutherfordium 104 **Rf** [265.12]	Dubnium 105 **Db** [268.13]	Seaborgium 106 **Sg** [271.13]	Bohrium 107 **Bh** [270]	Hassium 108 **Hs** [277.15]	Meitnerium 109 **Mt** [276.15]	Darmstadtium 110 **Ds** [281.16]	Roentgenium 111 **Rg** [280.16]	Copernicium 112 **Cn** [285.17]	Ununtrium 113 **Uut** [284.18]	Flerovium 114 **Fl** [289.19]	Ununpentium 115 **Uup** [288.19]	Livermorium 116 **Lv** [293]	Ununseptium 117 **Uus** [294]	Ununoctium 118 **Uuo** [294]

Key:

Element Name
Atomic number
Symbol
Atomic weight (mean relative mass)

*lanthanoids	Lanthanum 57 **La** 138.91	Cerium 58 **Ce** 140.12	Praseodymium 59 **Pr** 140.91	Neodymium 60 **Nd** 144.24	Promethium 61 **Pm** [144.91]	Samarium 62 **Sm** 150.36(2)	Europium 63 **Eu** 151.96	Gadolinium 64 **Gd** 157.25(3)	Terbium 65 **Tb** 158.93	Dysprosium 66 **Dy** 162.50	Holmium 67 **Ho** 164.93	Erbium 68 **Er** 167.26	Thulium 69 **Tm** 168.93	Ytterbium 70 **Yb** 173.05
**actinoids	Actinium 89 **Ac** [227.03]	Thorium 90 **Th** 232.04	Protactinium 91 **Pa** 231.04	Uranium 92 **U** 238.03	Neptunium 93 **Np** [237.05]	Plutonium 94 **Pu** [244.06]	Americium 95 **Am** [243.06]	Curium 96 **Cm** [247.07]	Berkelium 97 **Bk** [247.07]	Californium 98 **Cf** [251.08]	Einsteinium 99 **Es** [252.08]	Fermium 100 **Fm** [257.10]	Mendelevium 101 **Md** [258.10]	Nobelium 102 **No** [259.10]